

LUZERN, Switzerland

WMO: 066500

Lat: 47.0364N

Lon: 8.3008E

Elev: 456

StdP: 95.97

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.4	(c) -5.4	(d) -10.7	(e) 1.6	(f) -5.8	(g) -8.7	(h) 1.9	(i) -4.3	(j) 10.1	(k) 9.0	(l) 8.4	(m) 8.8	(n) 1.8	(o) 270	(p) 0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.8	(c) 30.3	(d) 20.4	(e) 28.5	(f) 19.9	(g) 26.7	(h) 19.1	(i) 21.2	(j) 28.4	(k) 20.4	(l) 27.2	(m) 19.6	(n) 25.7	(o) 2.2	(p) 170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.7	14.3	24.3	18.0	13.7	23.5	17.3	13.1	22.6	63.7	28.6	60.6	27.3	57.9	25.8	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.2	(b) 5.4	(c) 4.2	(e) -9.5	(f) 33.1	(g) 2.6	(h) 1.2	(i) -11.4	(j) 34.0	(k) -12.9	(l) 34.7	(m) -14.4	(n) 35.4	(o) -16.3	(p) 36.3		
(a) 7.2	(b) 5.4	(c) 4.2	(e) -10.0	(f) 23.1	(g) 2.6	(h) 0.9	(i) -11.9	(j) 23.7	(k) -13.4	(l) 24.2	(m) -14.9	(n) 24.7	(o) -16.8	(p) 25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	1.4	1.9	6.2	10.6	14.2	18.4	20.0	19.3	15.3	11.0	5.5	1.9
	DBStd	7.60	3.80	3.95	4.01	3.78	3.68	3.76	3.17	3.19	3.20	3.53	3.35	3.47
	HDD10.0	1089	268	227	128	38	7	1	0	0	2	32	138	249
	HDD18.3	3042	526	460	377	233	136	46	19	27	100	226	384	508
	CDD10.0	1280	1	0	9	56	136	251	311	288	161	64	4	0
	CDD18.3	190	0	0	0	0	7	47	71	56	9	0	0	0
	CDH23.3	1449	0	0	0	9	75	388	540	394	44	0	0	0
	CDH26.7	392	0	0	0	1	10	111	157	113	2	0	0	0
Wind	WSAvg	1.5	1.5	1.5	1.7	1.7	1.7	1.6	1.6	1.4	1.3	1.2	1.3	1.4
Precipitation	PrecAvg	1509	82	90	95	116	166	175	175	175	124	103	106	103
	PrecMax	1866	180	216	238	235	286	258	341	345	239	181	248	231
	PrecMin	1174	10	36	34	20	39	94	70	44	45	9	1	2
	PrecStd	173	46	45	47	54	65	44	68	69	47	41	54	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.4	14.3	19.2	24.8	28.0	32.1	32.6	32.5	26.6	21.5	16.3	11.9
		MCWB	9.7	8.9	11.0	14.3	17.9	21.0	21.1	20.3	19.2	15.8	11.6	8.4
	2%	DB	10.7	11.5	17.0	22.3	25.4	29.7	30.3	29.6	24.7	19.3	13.6	9.6
		MCWB	7.2	6.9	10.1	13.4	16.6	20.2	20.2	20.7	18.3	14.8	10.3	7.2
	5%	DB	8.3	9.4	15.0	20.0	23.3	27.6	28.4	27.4	22.5	17.6	11.6	8.0
		MCWB	5.9	5.8	9.3	12.3	15.5	19.3	19.7	20.0	17.3	13.9	9.4	6.3
	10%	DB	6.6	7.4	12.8	17.8	21.1	25.4	26.4	25.3	20.6	16.0	10.2	6.7
		MCWB	4.8	4.8	8.1	11.2	14.5	18.4	18.9	19.1	16.3	13.3	8.4	5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.0	9.0	11.8	15.2	18.9	22.4	22.2	22.1	20.1	16.7	12.1	9.0
		MCDB	14.5	13.2	17.5	23.1	26.5	29.9	30.5	29.0	25.2	20.3	15.3	10.9
	2%	WB	7.7	7.5	10.6	13.9	17.6	20.9	21.1	21.1	18.7	15.5	10.8	7.6
		MCDB	10.0	10.8	15.8	20.7	24.0	28.0	28.4	28.0	23.5	18.3	13.1	9.1
	5%	WB	6.2	6.1	9.6	12.7	16.2	19.9	20.2	20.3	17.7	14.5	9.7	6.5
		MCDB	8.2	9.1	14.3	19.0	21.6	26.3	27.1	26.7	21.8	16.9	11.3	7.8
	10%	WB	4.9	4.9	8.5	11.6	15.1	18.9	19.4	19.4	16.7	13.5	8.6	5.4
		MCDB	6.4	7.2	12.5	16.9	19.9	24.5	25.4	24.7	20.0	15.6	10.0	6.7
Mean Daily Temperature Range	5% DB	MDBR	5.1	6.5	8.8	9.9	9.4	9.6	9.8	9.2	8.2	7.0	5.6	4.7
		MCDBR	8.3	10.4	12.8	14.1	13.4	13.5	13.1	12.6	11.1	9.7	8.1	7.0
		MCWBR	5.6	6.5	7.1	6.9	6.5	6.0	5.6	5.6	5.7	5.8	5.2	4.8
	5% WB	MCDBR	8.0	9.9	11.8	13.1	11.9	12.3	12.0	11.9	10.2	8.8	7.7	6.6
		MCWBR	5.5	6.4	6.6	6.6	6.0	5.8	5.4	5.5	5.4	5.3	5.2	4.6
Clear-Sky Solar Irradiance	taub	0.293	0.324	0.372	0.414	0.432	0.437	0.438	0.426	0.395	0.365	0.324	0.290	
	taud	2.357	2.274	2.212	2.162	2.173	2.215	2.210	2.257	2.322	2.376	2.415	2.380	
	Ebn at Noon	792	833	841	838	835	832	824	818	810	778	748	751	
	Edh at Noon	72	97	120	139	143	138	137	126	107	88	69	63	
All-Sky Solar Radiation	RadAvg	1.33	2.22	3.42	4.38	4.92	5.73	5.61	4.88	3.73	2.40	1.39	1.10	
	RadStd	0.16	0.29	0.48	0.66	0.49	0.53	0.50	0.47	0.37	0.26	0.18	0.16	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	+0.34	N/A	N/A	N/A	N/A	N/A
(47) Regional (121 neighbors)	N/A	N/A	N/A	N/A	+0.72	+0.40	+0.31	N/A	N/A	+57	+15

Nomenclature: See separate page